

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (EC) Examination

May 2014

EC 309 Digital Communication

Date: 08.05.2014, Thursday

Time: 10:00 am To 01:00 p.m.

Maximum Marks: 70

Instructions:

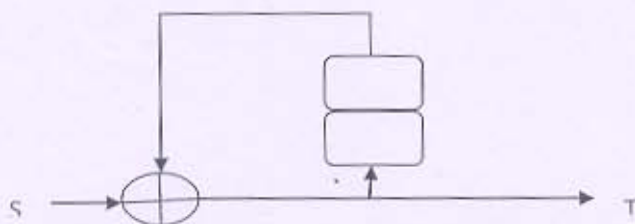
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) Explain the generation of binary frequency shift keying modulation technique. Draw the constellation diagram for same. [04]
- (b) Define bit rate and baud rate with example. [03]
- Q - 2 (a) Write short note on scrambling. [07]
- (b) Find a generator polynomial $g(x)$ for a (7,4) cyclic code, and find code vectors for the following data vectors: 1010, 1111. [07]

OR

- Q - 2 (a) A scrambler is shown in figure. Design the corresponding descrambler. If a sequence $S = 101010100000111$ is applied to the input of this scrambler, determine the output sequence T . verify that if this T is applied to the input of the descrambler, the output sequence S . [07]



- (b) Write short note on Viterbi's decoding algorithm. [07]
- Q - 3 (a) Give the comparison between coded and un coded systems. [07]
- (b) Define PDF and CDF, State the properties of CDF with proof. [07]

OR

- Q - 3 (a) Explain burst error detecting and correcting codes [07]
- (b) A Binary symmetric channel (BSC) error probability is P_e . The probability of transmitting 1 is Q , and that of transmitting 0 is $1-Q$. determine the probability of receiving 1 and 0 at the receiver. [07]

SECTION – II

Q - 4 (a) Explain the generation of binary amplitude shift keying modulation technique. Draw the constellation diagram for same. [04]

(b) What is the Nyquist criterion for zero ISI? [03]

Q - 5 (a) Write short note on Chebyshev's Inequality. [07]

(b) A zero memory source emits messages m_1 and m_2 with probabilities 0.8 and 0.2, respectively. Find the optimum (Huffman) binary code for this source as well as for its second order extension. Determine the code efficiencies in each case. [07]

OR

Q - 5 (a) Write short note on Central limit theorem. [07]

(b) Shows that entropy is maximum when all the messages are equiprobable. [07]

Q - 6 (a) Explain error free communication over noisy channel. [07]

(b) Write short note on sampling theorem. [07]

OR

Q - 6 (a) Find the channel capacity of BSC channel. Draw the graph of channel capacity v/s probability of error and give justification for same. [07]

Assume error probability is P_e , probability of transmitting 1 (x_1) is α and transmitting 0 (x_2) is $1 - \alpha$.

(b) Give the comparison between delta modulation and adaptive delta modulation. [07]
